

TrueYield recovery report — Clark County-NV - Government Center (system 1420)

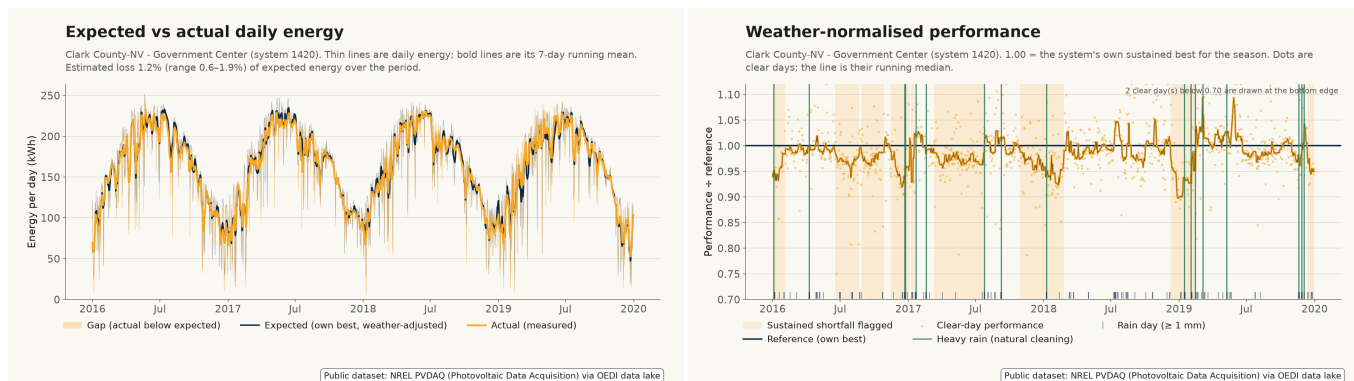
Data label: Public dataset: NREL PVDAQ (Photovoltaic Data Acquisition) via OEDI data lake | Period 2016-01-01 to 2019-12-31 | 30 kWp, tilt 10°, azimuth 180° | generated 2026-10-04

1. What is measured (facts)

- Generation: 1455 valid days of 1461 (99.6% complete); 2 days with no data, 0 partial days and 4 days with impossible values were excluded, not filled in.
- Loader inferred: pvdaq_system_1420.csv: system 1420; time column 'measured_on'; cumulative_counter from ['ac_kwhr_kwhac_4813'] in kWh; 15-minute steps. sampling interval varies: 15 min on 1008 days, 5 min on 451 days.
- Weather: NASA POWER satellite irradiance (all-sky and clear-sky), 2 m temperature and rainfall for the site coordinates. No on-site sensor is used.

2. What is estimated, and how sure

- Estimated loss: 1.2% of expected energy (range 0.6–1.9%), 2,849 kWh (range 1,438–4,484 kWh) over 1455 valid days. Of this, 2,849 kWh (range 1,438–4,484) is sustained shortfall in 9 flagged episode(s), and 0 kWh (range 0–0) is from days with near-zero output. Ranges are 95% bootstrap intervals covering day-to-day noise and the uncertainty of the reference. Shortfalls below the trigger are counted as zero, so this is a conservative figure.
- Detection limit: losses below about 4.1% sustained over 14 days, or about 4.0% sustained over 30 days, cannot be distinguished from noise (from clear-day scatter of 3.0% per day; 1% false-alarm rate per window, 80% power, starting from a system at its reference). Checked with injected-loss trials on real data from system 1420: reliable detection (at least 80% of trials) was reached at about 4% over 14 days, and at about 4% over 30 days. Smaller losses are counted as zero.
- Method: Hourly mode. NASA POWER hourly global irradiance is split into beam and diffuse (Erbs), transposed to the array plane (Hay-Davies, pvlib) and converted to DC energy with a PVWatts model (temperature coefficient -0.4%/°C, NOCT 45 °C). Hourly values are summed to days. Performance index = measured ÷ modelled energy. The reference is the system's own 90th-percentile sustained (7-clear-day median) performance per calendar month, pooled over all years and reduced by the upward bias expected from noise. Clear day = all-sky ÷ clear-sky irradiation ≥ 0.85. Assumptions: constant inverter and wiring losses are absorbed by the reference; timestamps are local standard time; the cleanest state seen in each month is representative of a clean array.



3. Likely causes: detected facts vs hypotheses

Cause	Detected facts	Hypothesis (not proven)
outage or fault not indicated	No day with good irradiance had near-zero output. 2 day(s) have no usable generation data (nothing logged, or a reading missing from at least one inverter); whether the system produced on those days is unknown and they are not counted as loss.	—
soiling not indicated	Between cleaning events, performance declined in 22 of 22 usable intervals, at a median of 0.179% per day in the declining ones. 0 of 16 testable natural rain event(s) were followed by a rise in normalised performance that is distinguishable from zero. rdtools estimates a soiling loss of 4.2% (range 2.9–6.1%, rdtools 3.2.1 stochastic rate and recovery (SRR)). Shuffle check: with the same clear-day values in random time order, which contain no soiling pattern, it reports 3.8–4.6%.	No soiling distinguishable from scatter. The estimate of 4.2% (range 2.9–6.1%, rdtools 3.2.1 stochastic rate and recovery (SRR)) is not clearly above what the same method reports on shuffled data (up to 4.6%), so most or all of it can be produced by day-to-day scatter alone.
ageing or degradation not indicated	Year-on-year change of clear-day performance over 4.0 years: +0.18% per year (range -0.31 to +0.59% per year, rdtools 3.2.1 year-on-year).	No ageing trend distinguishable from zero: the range includes zero or is positive.
shading cannot be assessed	Clock check: on clear days the measured daily profile is centred +37 minutes from the modelled one (+44 min in May–August, -6 min in November–February). The timestamps do not line up with solar time well enough for an hour-by-hour test (the summer–winter difference suggests the logger follows daylight-saving time).	—

4. Recovery verification

- Events tested: natural rain event, not a maintenance intervention. 16 of 17 event(s) had enough clear days on both sides (14 days before, 14 after).
- Best-observed event, 2017-07-25 (natural rain event, not a maintenance intervention): +4.0% (range -2.5 to +10.3%), +7.2 kWh/day (range -4.6 to +18.1), from 9 clear days before and 10 after. It was chosen for having the most clear days, not for its size. The range includes zero: no recovery can be claimed for this event.
- Across all testable events: 0 clear rise(s), 0 clear fall(s), 16 not distinguishable from zero. Full list in summary.json.

5. Value and verdict

- [not available: tariff_inr_per_kwh not provided in data/systems.csv]
- Emissions: [not available: emission_factor_kg_per_kwh is empty in config.yaml; no factor is assumed]

6. Limits of this report

- Causes are hypotheses; none is proven without a site inspection.
- Loss is measured against this system's own best month by month; a loss present from the start, or one that recurs in the same month every year, is partly or wholly invisible.
- Irradiance is a satellite area average; losses below the detection limit are not counted.

7. Method validation with injected loss — not a field result

- On a clean real stretch of this system (2018-04-13 to 2018-12-08), a 10% soiling ramp and a 3-day outage were injected (1,221 kWh). The unchanged pipeline recovered 1,041 kWh (85% of the injected loss; estimate 1,041 kWh, range 762–1,610); the known truth is inside the stated range. Outage days found: 3 of 3. Detection-limit check: over 14 days: 2% step found in 27%, 4% step found in 80%, 6% step found in 93%, 8% step found in 97%, 10% step found in 100%, 12% step found in 100%; over 30 days: 2% step found in 47%, 4% step found in 90%, 6% step found in 100%, 8% step found in 100%, 10% step found in 100%, 12% step found in 100% of 30 trials each. These are test results on altered data, not field results.